

Loss Given Default Backtesting under Discrete Rating Scales

Revisiting the Statistical Assumptions of Standard ECB Validation Procedures

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Internal Validation Function

- The regulatory framework for IRB models assigns a central role to the internal validation function in ensuring that models remain adequate, robust, and reliable throughout their lifecycle.
- Under the Capital Requirements Regulation (CRR), banks are required to maintain a comprehensive validation process that provides independent evidence that the internal estimates used for regulatory capital calculations remain appropriate.
- Validation typically serves as the first line of assurance for both senior management and supervisors, supporting decisions on model approval, ongoing monitoring, and model changes.
- When establishing their validation frameworks, banks, with the primary objective of complying with regulatory requirements, seek to standardize validation procedures, making them as consistent and comparable as possible across the entire model landscape.
- Moreover, banks thoroughly review and often seek to align their validation frameworks with the methods presented or promoted by supervisory authorities, generally considering them to represent best practice.
- One of the documents that banks frequently consult when performing validation activities and developing their validation frameworks is the [ECB Instructions for Reporting the Validation Results of Internal Models – IRB Pillar I Models for Credit Risk](#).

ECB Supervisory Monitoring

- To facilitate supervisory monitoring through a comparable quantitative framework, the ECB issued the [Instructions for Reporting the Validation Results of Internal Models – IRB Pillar I Models for Credit Risk](#).
- The document explains the rationale for the additional supervisory reporting and introduces a common set of validation tools, statistical tests, and analyses that enable the ECB to perform a more consistent initial assessment of model performance across institutions.
- The instructions explicitly acknowledge that they are not legally binding and do not replace or override applicable European or national legislation.
- Furthermore, the ECB emphasizes that the prescribed validation tools should not be regarded as the minimum set of analyses required to demonstrate regulatory compliance. Instead, institutions remain responsible for designing validation frameworks that are proportionate to the complexity, materiality, and intended use of their internal models.
- Nevertheless, institutions supervised by the ECB are expected to report the results of the prescribed validation tools as part of the additional supervisory information requested under individual supervisory decisions.

ECB Supervisory Monitoring cont.

- Regarding the interpretation of the submitted results, the ECB also recognizes that statistical validation techniques have inherent limitations and rely on assumptions that may not always hold in practice, particularly for small samples. However, no other explicit limitations are discussed.
- Given that banks typically regard ECB documents as one of the most influential references when designing their modeling frameworks, it is worth asking whether certain approaches presented in this document should be further challenged and their limitations more thoroughly explored.
- The purpose of this presentation is to challenge certain assumptions underlying the Loss Given Default (LGD) predictive ability testing framework for one of the commonly used discrete rating scale approaches. Specifically, the formulation of the null hypothesis is questioned from both a statistical perspective and the perspective of the underlying nature of IRB models.

ECB LGD Backtesting Using a T-test

In its instructions, the ECB defines the t-test as a one-sample test for paired observations, which compares estimated LGD with realised LGD under the null hypothesis that the estimated LGD is greater than the true LGD (one-sided hypothesis test), assuming independent observations.

The test statistic is given as follows:

$$T = \sqrt{N} \frac{\frac{1}{N} \sum_{i=1}^N (LGD_i^R - LGD_i^E)}{\sqrt{s_{LGD}^2}}$$
$$s_{LGD}^2 = \frac{\sum_{i=1}^N \left((LGD_i^R - LGD_i^E) - \frac{1}{N} \sum_{j=1}^N (LGD_j^R - LGD_j^E) \right)^2}{N - 1}$$

where:

- N is the number of facilities used for testing;
- LGD_i^E denotes the estimated LGD for facility i ;
- LGD_i^R denotes the realised LGD for facility i .

ECB LGD Backtesting Using a T-test cont.

Under the null hypothesis, the test statistic is asymptotically Student's t-distributed with $N - 1$ degrees of freedom, where N denotes the number of facilities in the backtesting sample.

The p-value is then calculated as $1 - S_{N-1}(T)$, where $S_{N-1}(T)$ is the cumulative distribution function of the Student's t-distribution evaluated at the test statistic T with $N - 1$ degrees of freedom.

According to the instructions, backtesting is expected to be performed at both of the following levels:

- 1 portfolio level;
- 2 facility grade, pool, or segment level.

Note

Although the presented approach refers to a one-sided version of the paired t-test, in practice, the same approach is also used for testing the so-called accuracy or precision of the best estimates. Therefore, most of the challenges described in this presentation are equally applicable to the two-sided version of the proposed test.

Challenges with the Proposed Testing Procedure

- Given the proposed approach and the fact that, for discrete rating scale facility grade or pool estimates, the estimates are single values (e.g., long-run averages or final estimates), an important question arises: what exactly is this procedure testing?
- Additionally, under the assumption that the validation sample LGD represents the true LGD and given the proposed null hypothesis, several straightforward questions arise:
 - How does this assumption align with the standard statistical hypothesis testing framework?
 - How does this assumption align with the principles used in other backtesting exercises?
 - How does this assumption align with the nature of IRB models?
- The following slides describe some of the main challenges to the ECB's assumptions underlying the proposed test. The main goal of challenging this proposed approach is to raise awareness of the possible consequences of the assumptions used in defining the testing framework, as well as to emphasize that a single approach is typically not sufficient to conduct a robust model evaluation and validation exercise.

What Exactly is the Proposed Hypothesis Testing?

- Understanding this question is central to most of the challenges associated with the proposed testing procedure.
- Looking at the proposed testing framework, practitioners effectively assume that the true LGD is the one observed in the validation sample.
- Moreover, for a discrete rating scale, where testing is performed at the facility grade or pool level, the model estimate is a single value. Consequently, the entire variability of the test statistic is attributed to the variability observed in the validation sample.
- Combining these two assumptions, the testing procedure effectively examines whether the calibrated (estimated) average LGD is statistically significantly different from (or, in the case of a one-sided test, lower than) the average LGD observed in the validation sample, given the variability observed in the validation sample.
- This is fundamentally different from the question addressed by the standard hypothesis testing framework, which asks: If the calibrated LGD distribution represents the true underlying process, how unusual are the observed validation results?
- To illustrate the difference between the proposed paired t-test approach and the principles of standard hypothesis testing, let us compare it with the well-known z-score used in Probability of Default (PD) modeling.

Alignment with the Principles Used in Backtesting

The test statistic of the z-score test is given as follows:

$$z_{score} = \frac{ODR - PD}{\sqrt{\frac{PD(1-PD)}{N}}}$$

where:

- ODR is the observed default rate;
- PD is the calibrated PD;
- N is the number of exposures.

Note:

- As can be seen from the z-score test statistic, the null hypothesis assumes that the calibrated PD represents the true default rate. This assumption clearly differs from the one underlying the paired t-test proposed for LGD backtesting.

Alignment with the Nature of IRB Models

- Considering the Through-the-Cycle (TTC) nature of IRB models and the fact that they are calibrated using longer time series of realised losses, differences in variability between the calibration and validation samples can have a significant impact on the outcome of the test.
- While the choice of whether variability is measured from the calibration or validation sample is one important aspect of the nature of IRB models, another obvious consequence of the proposed one-sided paired t-test relates to the conservative nature of IRB LGD models.
- Specifically, the final estimates that are typically backtested using the proposed t-test are conservative model estimates. In other words, the Downturn (DT) adjustment and the Margin of Conservatism (MoC) make the final LGD estimates appropriate for the purpose of calculating regulatory risk weights.
- Moreover, as part of the DT quantification exercise, practitioners compare the estimated DT adjustment with reference values, potentially introducing additional conservatism into the final LGD estimates.
- Given the conservative nature of the final LGD estimates, it is reasonable to question whether the one-sided formulation of the proposed test is appropriate for continuous validation. In other words, under normal macroeconomic conditions, how likely is it that the final LGD estimates are underestimated?
- As stated in the ECB instructions, this observation reinforces the importance of designing a comprehensive backtesting framework rather than treating the proposed testing procedure as either the minimum requirement or the only appropriate validation approach.

Conclusions

- The ECB validation instructions provide a useful and standardized framework for supervisory monitoring. However, they explicitly acknowledge that the prescribed statistical tests are neither legally binding nor sufficient, on their own, to demonstrate overall model validity.
- For discrete rating scale LGD models, the proposed paired t-test effectively assumes that the validation sample represents the true LGD, attributing the entire uncertainty to the validation sample while treating the calibrated estimate as fixed.
- This interpretation differs from the standard hypothesis testing framework, in which the null hypothesis assumes that the calibrated model represents the true underlying process and evaluates the probability of observing the validation results under that assumption. Moreover, the proposed t-test differs from the principles commonly applied in other backtesting exercises, such as the PD z-score test, where uncertainty is derived from the calibrated model.
- Considering the Through-the-Cycle (TTC) nature and the inherent conservatism of IRB LGD estimates, the one-sided formulation of the proposed test raises additional questions regarding its suitability for continuous validation.
- The choice of statistical assumptions directly affects the interpretation of predictive ability results. Consequently, the proposed paired t-test should be viewed as one component of a broader validation framework rather than as a standalone or best-practice approach for assessing LGD predictive ability.